

Where electromagnetically induced transparency lives, and where it inverts, in the nitrogen–vacancy spin- Λ channel

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(Dated: August 13, 2026)

Electromagnetically induced transparency (EIT) has been observed in nitrogen–vacancy (NV[−]) ensembles at low temperature, but no microscopic account exists of why it disappears on warming, how far external control can recover it, or what remains at room temperature. We resolve the weak-probe response of the NV[−] zero-phonon-line spin- Λ channel with the full nine-level Lindblad generator and classify every point of the temperature–transverse-field plane by three criteria applied together: the sign of the coherence-mediated contrast, the lineshape of the control-on absorption, and an Akaike model comparison against Autler–Townes splitting. The region supporting genuine transparency is a closed island. It requires a transverse field $B_{\perp} \gtrsim 0.15$ T to open the Raman pathway at all; it is bounded below near 22 K by an Autler–Townes crossover, and above near 90 K–95 K by phonon-driven collapse of the leading Raman path, which costs four and a half orders of magnitude of contrast between 30 K and 100 K. Above 103 [97, 108] K the response changes sign: the control field no longer induces transparency but induces absorption. Raising $B_{\perp}$ from 0.15 T to 0.5 T moves the boundary by less than 2 K, so the transverse field opens the channel without buying temperature. Converting to laboratory observables on a single detection chain, the sign reversal itself lies inside the detectable region — reachable in seconds of integration — whereas at room temperature the residual contrast of $\sim 10^{-9}$ is unreachable at any integration time once a realistic technical-noise floor is imposed. We also delimit where the reduced kernel widely used for such estimates may be trusted: it reproduces the boundary temperatures but not the contrast magnitudes, and it fails qualitatively, including in sign, below 40 K and above 90 K.

I. INTRODUCTION

Electromagnetically induced transparency renders an absorbing medium transparent to a weak probe by destructive interference of excitation pathways opened by a strong control field [1]. In atomic vapours the mechanism is textbook. In solids it is not: the same Λ -shaped level diagram can yield a deep transparency window in one material and nothing at all in another, because real emitters carry many excited branches, several dissipative channels, and material-specific selection rules that the three-level picture does not contain.

The negatively charged nitrogen–vacancy centre in diamond is the most thoroughly characterised of these systems [2]. Coherent population trapping of single NV spins [3] and EIT in NV ensembles [4] have both been demonstrated at cryogenic temperature, and the appeal of extending them upward is obvious: an all-optical, room-temperature coherent interface in a material that already supports room-temperature spin read-out. What has been missing is an account of *why* the low-temperature effect does not survive warming. The usual explanation — that dephasing grows and washes the feature out — is incomplete, because it predicts only a monotonic fading. It does not predict a lower temperature bound, it does not predict a threshold in transverse field, and it does not predict a change of sign. All three occur.

We compute the weak-probe response of the NV[−] zero-phonon-line spin- Λ channel without adiabatic elimination, using the full nine-level Lindblad generator, over the temperature–transverse-field plane. The organising

quantity is not the total susceptibility but the part of it carried by the targeted ground-state coherence, isolated by an algebraic counterfactual: the sector-resolved correction $\delta\chi_S$. This distinction is not cosmetic. A vanishing *total* susceptibility is the signature of ideal EIT, not of its absence, so a criterion built on the total response has the sign of the conclusion backwards. Section II makes the construction precise.

Three results follow. First, the transparency region is a closed island in the $(T, B_{\perp})$ plane rather than a half-space below some temperature (Sec. III). Second, the transverse field is a switch and not a dial: it must exceed a threshold before there is any channel, and once open, further field buys essentially no temperature headroom (Sec. IV). Third, above ~ 103 K the coherence-mediated response inverts, so that the control field increases the probe absorption; this inversion is not a vanishing remnant but a signal that a realistic experiment can resolve in seconds, whereas the room-temperature residual cannot be resolved at all (Sec. VI).

Because a claim of this kind stands or falls on the validity of the model that produced it, Sec. VII reports the audit in full, including where the reduced description that is commonly used for such estimates — and that we ourselves use for the uncertainty propagation — may and may not be trusted. The general classification of dissipative response orders that the present analysis instantiates is developed separately and is not assumed here.

II. RESPONSE THEORY AND THE OBJECT THAT IS CLASSIFIED

A. Model

The NV^- ground manifold is the 3A_2 spin triplet with zero-field splitting $D_{gs} = 2.877$ GHz; the optically connected excited manifold is the six-dimensional 3E orbital doublet, carrying spin-orbit, spin-spin, transverse strain and Zeeman terms [5, 6]. We retain both manifolds in full, giving nine levels once the spectator ground sublevel is included, and evolve them with a time-independent Lindblad generator in the rotating frame,

$$\dot{\rho} = \mathcal{L}(\rho) = -i[H, \rho] + \sum_{\mu} \left(L_{\mu} \rho L_{\mu}^{\dagger} - \frac{1}{2} \{L_{\mu}^{\dagger} L_{\mu}, \rho\} \right). \quad (1)$$

The jump operators are spin-conserving radiative decay from each excited branch, phonon-induced orbital hopping $E_x \leftrightarrow E_y$ at rate $k_{\text{orb}}(T)$, ground-state relaxation, and ground-state dephasing. Intersystem crossing to the singlet and ^{14}N hyperfine structure are available as structural toggles and are used in the audit of Sec. VII.

The temperature enters through $k_{\text{orb}}(T)$, taken from the measured photophysics of single NV centres [7] and cross-checked against three alternative rate models as a systematic (Sec. V). The essential point for what follows is a property of the *coherence* block rather than of populations: a jump confined to the excited manifold, which merely shuffles population between E_x and E_y and removes none, nevertheless acts on every optical coherence as pure loss. Writing $\sigma = P_e \rho P_g$ for the optical-coherence block,

$$P_e \mathcal{D}[L_{\mu}](\rho) P_g = -\frac{1}{2} L_{\mu}^{\dagger} L_{\mu} \sigma, \quad L_{\mu} = P_e L_{\mu} P_e, \quad (2)$$

because the recycling term cannot connect the rectangular block to itself. Orbital hopping is therefore a decoherence channel for the Raman pathway even though it conserves excited-state population, and this is the microscopic origin of the collapse reported below. One consequence deserves emphasis because it is a recurring source of factor errors: for symmetric two-way hopping at rate k the population imbalance relaxes at $\Gamma_{XY} = 2k$, while each optical coherence is damped at $k/2 = \Gamma_{XY}/4$. The rate that enters Eq. (2) is a quarter of the orbital relaxation rate usually quoted.

B. The sector-resolved correction

Let the first-order response be organised into the optical-coherence vector x and the targeted ground coherence $s = \rho_{g_2 g_1}^{(1)}$,

$$\begin{pmatrix} A & B \\ C & G_g \end{pmatrix} \begin{pmatrix} x \\ s \end{pmatrix} = \begin{pmatrix} b_p \\ 0 \end{pmatrix}. \quad (3)$$

Eliminating s without ever dividing by G_g — so that the ideal ground-coherence limit is retained — gives the

probe response as a Schur complement. We define the *pathway-cut* counterfactual $\chi_{\text{cut}}^{(S)}$ by setting $B = C = 0$ while holding A , G_g , the source, the detector and the zeroth-order state fixed, and the sector-resolved correction

$$\delta\chi_S \equiv \chi_{\text{full}} - \chi_{\text{cut}}^{(S)} = \chi_0 d^{\dagger} A^{-1} B S_g^{-1} C A^{-1} b_p, \quad (4)$$

with $S_g = G_g - C A^{-1} B$. Throughout we report the dimensionless contrast

$$C = \frac{A_{\text{cut}} - A_{\text{full}}}{A_{\text{cut}}}, \quad (5)$$

where A_{full} and A_{cut} are the control-on absorptions with and without the ground-coherence pathway. With this convention $C > 0$ means the pathway *reduces* absorption — transparency — and $C < 0$ means it increases it.

Two properties of Eq. (4) matter for the interpretation of every figure below. It is an algebraic counterfactual, not a control-off measurement: switching the control off would also change optical pumping, light shifts and the steady state, and would therefore not isolate the coherence pathway. And it is the correct object to test, because $\chi_{\text{full}} = 0$ is what *perfect* EIT looks like. In the ideal three-level limit the total response vanishes while $\delta\chi_S$ is maximal; a criterion phrased on the total susceptibility would report that case as an absence of EIT.

C. Classification

At each point of the $(T, B_{\perp})$ plane we scan the two-photon detuning with an adaptive window — the feature width varies over three orders of magnitude across the plane — and assign one of four labels using the sign of the peak contrast, the containment of the feature, and a model comparison of the control-on lineshape. The lineshape test follows Anisimov, Dowling and Sanders [8]: the absorption profile is fitted with an EIT (interference) model and an Autler–Townes (doublet) model, together with Lorentzian and Fano alternatives, and the verdict is fixed by $\Delta\text{AIC} = \text{IC}_{\text{ATS}} - \text{IC}_{\text{EIT}}$ with a robust gate at $|\Delta\text{AIC}| \geq 6$. A point is labelled *genuine transparency* only if $C > 0$, the feature is contained in its window, $|C|$ exceeds the detection floor, and the lineshape is robustly EIT. A negative C is labelled *control-induced absorption*; a robustly ATS lineshape is labelled Autler–Townes; everything else is *unresolved*, which includes both sub-threshold features and genuinely inconclusive lineshapes.

Requiring all three criteria jointly is what separates this classification from a contrast map. A large $|C|$ alone is not evidence of transparency: C is a ratio, and where the probe leg is nearly closed the denominator is small, so $|C|$ can be large while the absolute absorption change is negligible — or, as we find at low field, while the control is in fact turning absorption *on*. We therefore record the absolute absorptions alongside C everywhere.

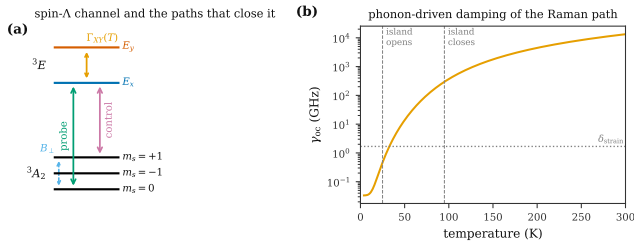


FIG. 1. (a) The NV^- spin- Λ channel. Probe and control address two ground spin sublevels through the 3E orbital doublet. Because the optical dipole is orbital, the two legs reach orthogonal spin states and the leading Raman vertex is closed at $B_\perp = 0$; a transverse field mixes the ground sublevels and opens it. Phonon-induced hopping $\Gamma_{XY}(T)$ between the excited branches damps the pathway. (b) Optical-coherence damping $\gamma_{oc}(T)$ over the range studied, with the transverse strain splitting marked for scale. The island boundaries found in Sec. III are indicated.

III. THE TRANSPARENCY ISLAND

Figure 3 is the central result. Two features of it are not anticipated by a dephasing-washout picture.

A threshold in transverse field. At $B_\perp = 0$ there is no usable transparency at any temperature. This is not an artefact of small numbers: the optical dipole is orbital, so with unmixed ground sublevels the two legs of the Λ terminate on orthogonal spin states and the leading Raman vertex vanishes identically. What survives at zero field is a small residual pathway through spin mixing inside 3E , of order 10^{-5} in contrast at 70 K, which is a different pathway and which we separate explicitly in Sec. IV. Genuine transparency appears only for $B_\perp \gtrsim 0.15$ T.

A lower temperature bound. Below about 22 K the region closes again. As the optical line narrows on cooling, the fixed control Rabi frequency $\Omega_c = 0.1$ GHz becomes large compared with the optical-coherence damping, and the response crosses into the Autler–Townes regime: at 20 K the lineshape test returns robust ATS at the larger fields and the contrast is negative at the smaller ones, with a wide feature (FWHM ~ 70 MHz at 10 K) that is a dressed-state doublet rather than an interference window. The transparency region is therefore an island, bounded below by a control-power crossover and above by dissipation.

Between those bounds the collapse with temperature is steep. At $B_\perp = 0.232$ T the contrast falls from 0.99 at 30 K through 0.58 at 50 K, 1.4×10^{-2} at 70 K and 1.5×10^{-4} at 90 K to 2.1×10^{-5} at 100 K — four and a half orders of magnitude over 70 K — and then changes sign. The upper edge of the transparency label sits at 90 K for $B_\perp \leq 0.25$ T and 95 K above it, with a band of lineshape-inconclusive points immediately beyond, so the boundary is not sharp; the sign reversal that follows is.

A. Why the sign reverses: a damping-induced phase slip

The sign of $\delta\chi_S$ is set by the phase of the Raman vertex product $K_{12}K_{21}$ relative to the ground-coherence denominator $\gamma_g + \beta S_2$ in Eq. (4): writing $K_{12}K_{21}/(\gamma_g + \beta S_2) = r e^{i\phi}$ with $r > 0$, the correction is $\delta\chi_S \propto -r \cos \phi$, so the pathway is transparency-producing while $|\phi| < 90^\circ$ and absorption-producing once $|\phi|$ exceeds it. At the candidate field, $\arg(K_{12}K_{21})$ runs from -0.1° at 30 K through -17.0° at 70 K and -62.7° at 90 K to -89.3° at 100 K, while the denominator’s phase stays small ($\lesssim 15^\circ$ throughout); the combined phase ϕ crosses -90° between 100 K and 103 K, exactly where the numerically observed sign reversal sits.

The mechanism is the phase lag of a damped driven resonance. K_{12} and K_{21} are matrix elements of the resolvent $G = [\gamma_{oc}(T)I + i2\pi(H - zI)]^{-1}$: while the optical-coherence damping $\gamma_{oc}(T)$ is small compared with the excited-state splittings, G responds essentially in phase with the drive and the Raman pathway interferes *destructively* with the direct absorption path, which is transparency. As $\gamma_{oc}(T)$ grows past that internal scale — here the transverse-strain splitting $\delta \sim 1.7$ GHz, reached once phonon-induced orbital hopping brings $\gamma_{oc}(T) \gtrsim \delta$ near 100 K — the resolvent lags the drive by more than a quarter cycle, and the same interference term that used to cancel absorption now reinforces it: destructive interference has become constructive, i.e. the coherent pathway behaves as electromagnetically induced absorption rather than transparency. No new parameter or mechanism is introduced; the reversal is the phase response of a resonance being pushed through overdamping by the same temperature-dependent rate that drives the amplitude collapse of Sec. III, which is why the two occur within one narrow temperature interval rather than as independent features.

IV. HOW THE TRANSVERSE FIELD OPENS THE PATHWAY

That a transverse field opens the channel is expected on symmetry grounds: $B_\perp$ mixes the ground spin sublevels and thereby supplies the spin overlap that the orbital dipole cannot. Because the mixing enters once in each leg of the Λ , and the sector-mediated response is bilinear in the two Raman vertices, the leading behaviour should be $C \simeq C_{\text{res}} + aB_\perp^2$, with C_{res} the field-independent residual pathway through 3E spin mixing.

The field range over which that law can be tested is bounded from both sides, and narrowly. From above, the mixing ceases to be perturbative once the ground Zeeman energy reaches the zero-field splitting, $\gamma_e B_\perp = D_{gs}$ at $B_\perp = 0.103$ T; we take 0.031 T, three tenths of that, as the ceiling of the perturbative window. From below, the fit needs the opened part to stand clear of C_{res} , which happens near 0.008 T at these temperatures.

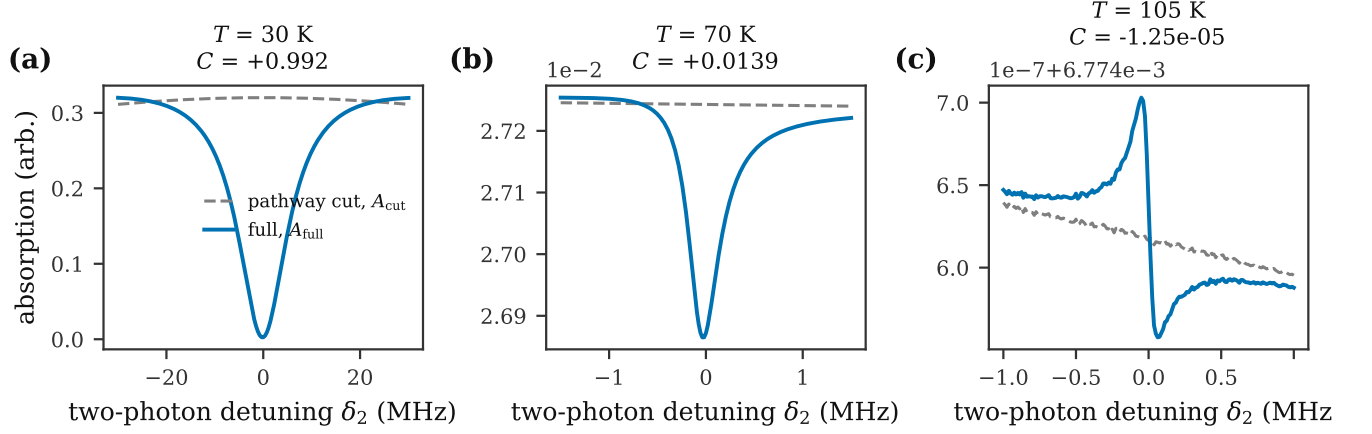


FIG. 2. Full-Liouvillian absorption spectra at the candidate field $B_{\perp} = 0.232$ T, with the ground-coherence pathway intact (A_{full} , solid) and cut (A_{cut} , dashed). (a) At 30 K the pathway removes essentially all of the sector absorption. (b) At 70 K a per-cent-level transparency survives. (c) At 105 K the two curves have crossed: the pathway now *adds* absorption, and the contrast has changed sign.

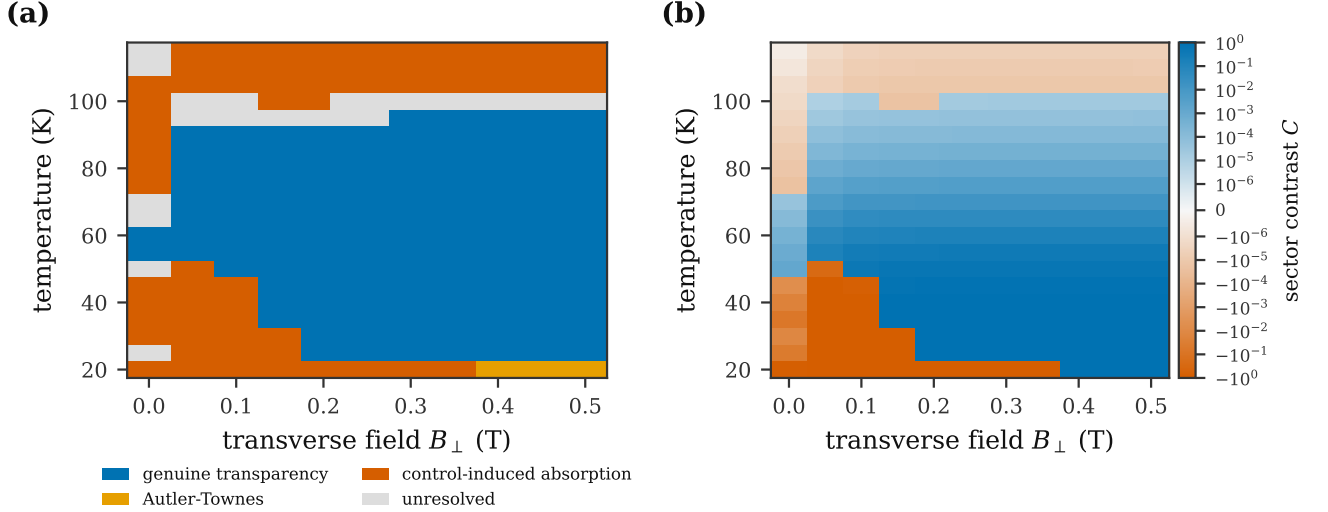


FIG. 3. (a) Classification of the $(T, B_{\perp})$ plane from the full nine-level Liouvillian, using sign, lineshape containment and the EIT/ATS model comparison jointly. Genuine transparency occupies a closed island: it requires $B_{\perp} \gtrsim 0.15$ T, closes below ~ 22 K through an Autler–Townes crossover, and closes above 90 K–95 K through phonon-driven collapse. (b) The same plane on a single signed scale: blue is $C > 0$ (transparency), red is $C < 0$ (control-induced absorption), and colour depth is $|C|$ on a symmetric-log axis. Fixed parameters: $B_z = 5$ mT, $\Omega_c = 0.1$ GHz, transverse strain $\delta = 1.683$ GHz.

Less than a decade of field separates the two. Note also that C_{res} cannot be read off at $B_{\perp} = 0$: that point is structurally degenerate and does not join smoothly onto the small-field plateau, so the residual has to be a fitted parameter, which is what including the plateau points in the fit constrains.

Fitting $C = C_{\text{res}} + aB_{\perp}^n$ on a dense grid inside that window separates the temperatures cleanly, and we report the separation rather than an average over it. At

85 K the exponent is stable against the choice of upper cutoff — it varies by 0.25 over cutoffs from 0.022 T to 0.046 T, with no monotone trend (Spearman $\rho = -0.05$) — and equals

$$n(85 \text{ K}) = 2.11 \pm 0.08, \quad (6)$$

which is the expected quadratic opening. At 70 K and 55 K it is not stable: the fitted exponent drifts monotonically downward with the cutoff ($\rho = -0.97$ at both, span-

ning 0.65 and 0.86 respectively), running from 2.7–3.0 at the smallest usable cutoff through 2 near $B_{\perp} \approx 0.04$ T. A genuine power law shows a plateau against the fitting window; a crossover shows exactly this drift. Figure 4(b) displays the distinction directly. We therefore claim the quadratic law at 85 K and above, and state that it is not resolvable at the cold end of the island.

Two controls establish that the drift belongs to the physics of the pathway and not to the way it is interrogated. Varying the control Rabi frequency over a factor of ten, $\Omega_c = 0.03$ to 0.3 GHz, moves the 70 K exponent from 2.664 to 2.628 — that is, not at all beyond the fit error. And fitting the absolute signal $\Delta A = A_{\text{cut}} - A_{\text{full}}$, which carries no field-dependent denominator, instead of the ratio C returns the same exponent (2.70 against 2.63 at 70 K), so the excess above 2 is not an artefact of normalising by a quantity that itself varies with field. Together with the archived room-temperature campaign, which returns $n = 2.03$ – 2.18 where the dynamic range is largest, the picture is that the opening is quadratic in its asymptotic regime and that the cold end of the island simply does not reach that regime before the mixing saturates.

The practical consequence does not depend on the exponent. $B_{\perp}$ behaves as a switch rather than a dial: Sec. V shows that raising it from 0.15 T to 0.5 T moves the temperature boundary by under 2 K. The field is necessary to have a channel at all, and once the channel is open the boundary is set by dissipation. This is the sense in which the transverse-field route is not a recovery mechanism for high-temperature EIT: it changes the prefactor of a suppressed response, not the temperature at which the suppression sets in.

V. WHERE THE BOUNDARY LIES, AND HOW WELL IT IS KNOWN

A boundary temperature quoted as a single number would misrepresent what is known. We propagate the experimental and model uncertainty by Monte Carlo: log-normal priors on the orbital-hopping and radiative rate scales, a normal prior on the transverse strain, priors on both field components, a log-uniform prior on the ground decoherence, a prior on the control Rabi frequency, polarization-angle errors, and — separately from the rate scales, which is where earlier treatments folded it — a log-uniform prior on an additional optical linewidth representing spectral diffusion. The four phonon-rate models enter as a discrete systematic rather than a single extrapolation.

Two thresholds are reported: $T_{1\%}$, where the contrast falls through 10^{-2} , and T_{sign} , where it changes sign. At the candidate field, from the full Liouvillian,

$$T_{1\%} = 74 [70, 78] \text{ K}, \quad (7)$$

$$T_{\text{sign}} = 103 [97, 108] \text{ K}, \quad (8)$$

as 68% intervals. The field dependence separates sharply into two regimes. For $B_{\perp} \leq 0.10$ T both thresholds collapse to 27 K–33 K and coincide with each other, meaning there is no usable window at all. For $B_{\perp} \geq 0.15$ T they separate and then saturate: $T_{1\%}$ moves from 70.3 K to 69.7 K and T_{sign} from 98.5 K to 103.8 K as the field goes from 0.15 T to 0.50 T. The dominant uncertainties are the transverse strain and the orbital-hopping rate scale; the spread across the four phonon models is 0.8 K–2.7 K, comfortably inside the ~ 9 K statistical band, so the boundary does not rest on a single extrapolation of the high-temperature rate.

These bands are computed with the reduced kernel, which is affordable at 5×10^2 samples. Since Sec. VII finds that kernel unreliable above 90 K — exactly where T_{sign} sits — we repeated the propagation at the candidate field with the full nine-level Liouvillian and the identical priors, at 80 samples. The bands overlap, with the full-model medians about 3 K higher: $T_{1\%} = 74.3 [69.5, 78.2]$ K against 71.1 [67.2, 75.7], and $T_{\text{sign}} = 103.2 [96.6, 108.2]$ K against 100.3 [94.2, 107.7]. We quote the full-model values throughout. The reduced kernel misplaces the contrast *magnitude* badly at these temperatures while still crossing the thresholds within its own stated uncertainty, which is why the bands survive at all — and why none of the magnitudes in Fig. 3 comes from it.

VI. WHAT AN EXPERIMENT WOULD SEE

Contrast is not an observable. Converting it into one requires a detection chain, and the conversion must use the same chain at every temperature or the comparison measures the sample rather than the physics. This is a real hazard here: the sector cross-section grows as the optical line narrows, so a fixed sample that is nearly transparent at room temperature ($\text{OD} \approx 0.04$) is opaque at 10 K ($\text{OD} \approx 16$). We therefore report the optical-depth-matched sample, $\text{OD}_{\text{sector}} = 1$, as the common axis, and record the required sample length alongside.

We convert to two readouts. Transmission gives $\Delta T/T = \expm1(\Delta\text{OD})$ with $\Delta\text{OD} = \text{OD}_{\text{sector}} C$. Fluorescence, which for NV ensembles is often the more practical channel and which has not previously been evaluated for this system, follows from the change in absorbed power; in the optically thin limit $\Delta\text{PL}/\text{PL} \rightarrow -C$, so a transparency appears as a fluorescence *dip* of the same fractional size. Both are propagated to the integration time needed for a signal-to-noise ratio of five, with shot noise and a relative technical-noise floor of 10^{-6} .

The outcome, Fig. 6, is that the interesting physics is observable and the room-temperature limit is not. At 70 K the transmission signal is 1.4×10^{-2} and needs microseconds. At 90 K it is 1.5×10^{-4} and needs 10 ms. At 105 K, past the sign reversal, it is -1.2×10^{-5} and needs 1.6 s; at 110 K, 5.6 s. The predicted inversion is thus not a formal remark about a vanishing quantity but a mea-

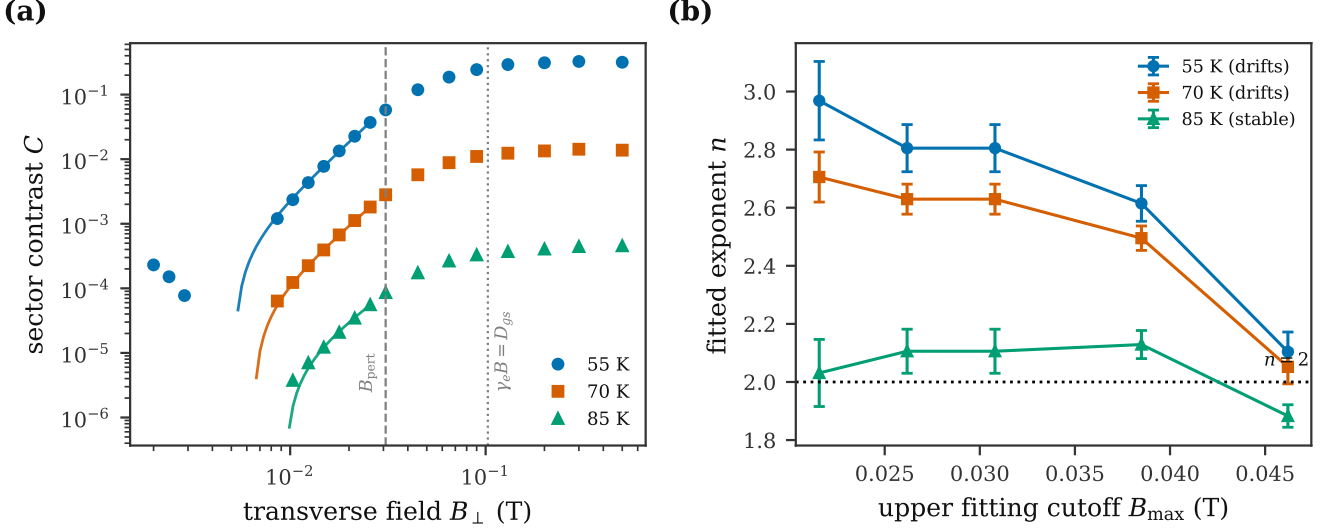


FIG. 4. Opening of the Raman pathway with transverse field, full Liouvillian. (a) Sector contrast against $B_{\perp}$; solid curves are fits of $C = C_{\text{res}} + aB_{\perp}^n$ inside the perturbative window, whose ceiling B_{pert} and the full-mixing field $\gamma_e B_{\perp} = D_{gs}$ are marked. (b) The fitted exponent against the upper fitting cutoff. A power law appears as a plateau and a crossover as a drift: 85 K plateaus at the expected $n = 2$, while 70 K and 55 K drift monotonically and do not determine an exponent.

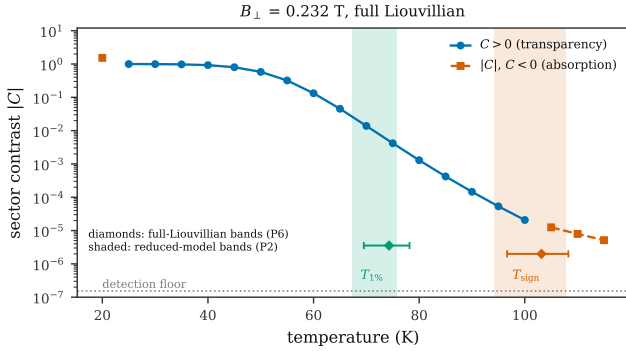


FIG. 5. Signed contrast against temperature at $B_{\perp} = 0.232$ T from the full Liouvillian; filled circles are $C > 0$, open squares $|C|$ with $C < 0$. Shaded bands are the reduced-model 68% intervals for $T_{1\%}$ and T_{sign} ; diamonds are the corresponding full-Liouvillian intervals. The dotted line is the contrast detection floor.

surement of a few seconds, with the sign of the fringe as the signature. Beyond 120 K the technical-noise floor makes the signal unreachable at any integration time, and at 300 K the residual contrast of 1.1×10^{-9} is nine orders of magnitude below what the chain can resolve. Fluorescence tracks transmission up to 105 K and then falls away first, because a sample thick enough to absorb strongly is also one whose fluorescence saturates against further changes in absorption.

VII. MODEL AUDIT

Reduced versus full description. Much of the literature estimate for systems of this kind, and our own uncertainty propagation, uses a reduced kernel in which the excited manifold is handled through a resolvent rather than by solving the full Liouvillian. Comparing the two on 108 points of the $(T, B_{\perp})$ plane, they agree to within 10% at 47 of them and disagree in sign at 13. The disagreements are structured, not random: they concentrate below 40 K, where the reduced kernel reports transparency ($C \approx +0.99$ at 20 K) while the full Liouvillian reports control-induced absorption ($C \approx -1.5$), and above 90 K, where no field gives agreement within 10%. The reduced kernel is accordingly used in this work for one purpose only — the threshold locations of Sec. V, which we verified independently against the full model — and every contrast magnitude reported here is from the full Liouvillian.

Truncation. Adding the singlet as a tenth level, or averaging over the ^{14}N hyperfine manifold, changes the peak contrast by at most 30% across 25 K–105 K and never changes its sign. Contrast magnitudes should therefore be read as carrying a structural systematic of this size; boundary temperatures and the sign reversal are unaffected.

Classification stability. At every temperature tested the EIT/ATS verdict is unchanged under 60 noise bootstraps at 2% of the feature depth and 30 randomized fit initializations, and under window variations of $0.5\times$ to $2\times$ in all but one case (90 K, 3/4).

Conventions and provenance. The sign convention is

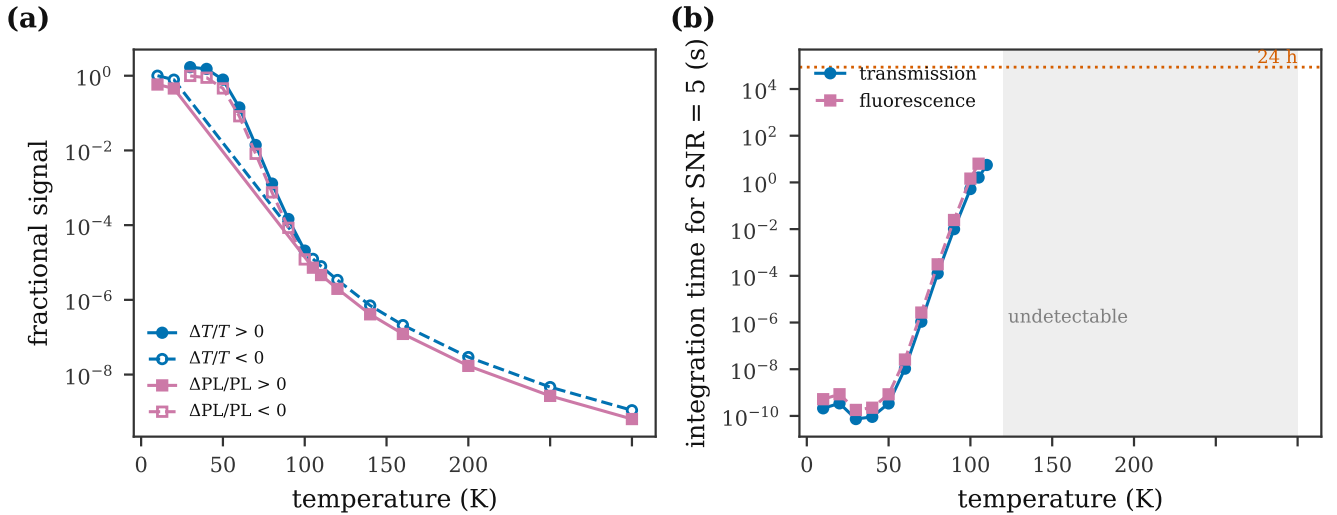


FIG. 6. (a) Fractional transmission and fluorescence signals against temperature for the optical-depth-matched sample; filled symbols positive, open symbols negative. (b) Integration time for SNR = 5 in each readout, with the 24 h ceiling marked. The sign reversal near 103 K lies inside the detectable region; the room-temperature residual does not.

fixed by Eq. (5) and verified directly at the candidate: $C > 0$ coincides with $A_{\text{full}} < A_{\text{cut}}$. Rate conventions follow the $\Gamma_{XY}/4$ mapping of Eq. (2). Literature parameters carry recorded sources and checksums.

Scope. The analysis is Markovian, weak-probe, and assumes a unique steady state. Non-Markovian spectral diffusion enters only through the effective linewidth prior of Sec. V, not as a bath with memory. Propagation and collective effects are outside the treatment.

VIII. DISCUSSION

The picture that emerges is more constrained than a temperature ceiling. NV spin- Λ EIT occupies a bounded island, and each of its three edges has a different mechanism: a symmetry obstruction that a transverse field must lift before there is any channel at all, a control-power crossover into Autler-Townes splitting at the cold edge, and phonon-driven collapse of the Raman path at the warm edge. Only the last of these is what the conventional dephasing argument describes.

This also sharpens what would count as a decisive experiment. A transparency dip at a single temperature and field establishes little, since a dip is equally the signature of Autler-Townes splitting; the discriminating measurements are the quadratic opening with transverse field at the warm end of the island, the collapse of the contrast with temperature over several decades, and above all the change of sign near 103 K, which no Autler-Townes mechanism produces and which we find to be resolvable in seconds. A measurement that followed the fringe through zero and out the other side would test the pathway-interference account directly.

For room temperature the conclusion is negative and, we believe, robust: the residual contrast is $\sim 10^{-9}$, nine orders below the detection floor of a chain that is already optimistic, and it is of the wrong sign in any case. Reaching a room-temperature solid-state EIT will require a configuration whose Raman pathway is not damped by an orbital degree of freedom that thermalizes at these rates — either a different defect, or NV under conditions that freeze the orbital dynamics out.

Group-IV centres are the obvious comparison, since their Λ channel is orbital rather than spin and so carries no spin-overlap obstruction [9–11]; their difficulty is the opposite one, an orbital ground coherence damped by single-phonon transitions [12], which strain engineering and larger spin-orbit splitting are designed to address [13, 14]. We treat that comparison as a design principle rather than a quantitative claim; the material-independent classification of response orders that would place NV and group-IV in a common scheme is developed elsewhere.

IX. CONCLUSION

We have located the region of the temperature-transverse-field plane in which the NV[−] spin- Λ channel supports genuine electromagnetically induced transparency, using the full nine-level Lindblad generator and classifying by sign, lineshape and an EIT/ATS model comparison together. The region is a closed island requiring $B_{\perp} \gtrsim 0.15$ T, bounded below near 22 K and above near 90 K–95 K, with the contrast falling four and a half orders of magnitude across it. Above 103 [97, 108] K the response inverts to control-induced absorption, an

inversion that a realistic measurement can resolve in seconds and that constitutes a direct test of the pathway-interference mechanism. At room temperature the residual response is unobservable by nine orders of magnitude. We have also delimited the validity of the reduced description commonly used for such estimates, which reproduces boundary temperatures but not contrast mag-

nitudes and fails qualitatively outside 40 K–90 K.

ACKNOWLEDGMENTS

The author thanks the members of the group for discussions.

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